

PROJECT REPORT

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Course Name: Electronic Circuits

Project Title:
Design, Implementation and Performance Study of a Power Supply

Submitted by

Name	Student ID
Naheda Islam Ela	2024-3-60-080
Abdullah Al Junaid	2024-3-60-082
Jumana Haider	2024-3-60-594
Eedy Kabir	2025-1-50-051

Submitted to

Dr. Md. Habibur Rahman
Professor, Department of Electrical and Electronic Engineering
University of Dhaka

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Group: 10A

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Abstract

This report documents the design, construction, and experimental evaluation of a linear, transformer-based, Zener-regulated DC power supply built for the CSE 251 Electronic Circuits course project. The circuit steps a 220V, 50Hz AC mains supply down through a center-tapped transformer, rectifies both half-cycles using a two-diode full-wave rectifier, smooths the resulting pulsating waveform with a 1000 μF electrolytic filter capacitor, and finally stabilizes the output at a nominal 6.2V using a Zener diode reference biased through a 200 Ω series resistor. The completed circuit was loaded with combinations of 1k Ω , 2.2k Ω , 4.7k Ω and 10k Ω resistors and six indicator LEDs to verify both its current-driving capability and its load-regulation performance. Across the full tested range from no load up to a load current of about 12.17mA the regulated output dropped by only approximately 0.72%, confirming that the design meets the fundamental requirement of a stable, low-ripple DC source suitable for driving small-signal digital and analog electronics.

Introduction

In modern electronics, a regulated direct current (DC) power supply is a fundamental necessity, as nearly all electronic devices require a smooth and constant DC voltage to operate correctly. Alternating current (AC) from the mains supply must undergo a systematic conversion process stepping down, rectification, filtering, and regulation to become usable and safe for sensitive semiconductor components such as microcontrollers, operational amplifiers, and digital logic gates.

Every consumer electronic device, from a simple LED nightlight to a fully-fledged embedded computer, ultimately depends on a power supply that can convert the raw, high-voltage AC delivered by the utility grid into a clean, low-voltage DC rail. Batteries offer one such DC source, but for mains-powered equipment, a dedicated AC-to-DC conversion circuit is indispensable. Understanding how each stage of that conversion chain works and, just as importantly, how each stage's imperfections propagate to the next is therefore one of the most practically useful skills an electronics student can develop.

A regulated DC power supply is a fundamental requirement in electronics, as the stable operation of logic circuits, microcontrollers, and operational amplifiers depends heavily on a ripple-free DC source. The primary objective of this project is to design, construct, and evaluate a linear regulated power supply capable of delivering a steady DC voltage despite load variations. The implemented circuit relies on a step-down center-tapped transformer, a two-diode full-wave rectifier, a high-

capacity electrolytic filter capacitor, and a Zener diode voltage regulator. The practical load-regulation characteristics were measured and analyzed to verify the circuit's stability.

This project focuses on the design, implementation, and performance evaluation of a linear regulated power supply capable of delivering a constant DC output regardless of fluctuations in the input mains or changes in the connected load. Beyond the immediate goal of producing a working circuit, the exercise also builds practical familiarity with component selection, breadboard prototyping, multimeter- and oscilloscope-based measurement, and the comparison of theoretical predictions against real, imperfect laboratory data skills that carry directly into more advanced power-electronics coursework and projects.

Objectives

- To systematically convert a 220V AC mains supply into a safe, regulated DC voltage.
- To implement a step-down, center-tapped transformer-based full-wave rectifier and analyze its rectification efficiency compared to half-wave alternatives.
- To utilize a 1000 μF smoothing capacitor to minimize the AC ripple factor in the pulsating DC output.
- To establish a stable voltage reference of $V_z = 6.2\text{V}$ using a Zener diode operating in its reverse breakdown region.
- To size the series resistor (R_1) correctly so that the Zener diode is held within its safe operating current range across the full expected load range.
- To experimentally determine the load regulation of the power supply by varying the load resistance (R_L) from 500Ω (two $1\text{k}\Omega$ resistors in parallel) up to open-circuit (no-load) conditions, using the $1\text{k}\Omega$, $2.2\text{k}\Omega$, $4.7\text{k}\Omega$ and $10\text{k}\Omega$ resistors provided.
- To drive six indicator LEDs, each with its own current-limiting resistor, from the regulated output and confirm flicker-free, stable illumination.
- To compare the measured load-regulation performance against the theoretical predictions and identify the practical sources of deviation.

Theory and Experimental Methods

A linear regulated power supply consists of four distinct operational stages: step-down, rectification, filtration, and regulation. Each stage addresses a specific shortcoming of the raw mains supply the transformer solves the voltage-level and safety-isolation problem, the rectifier solves the polarity problem, the filter solves the ripple problem, and the regulator solves the load- and line-dependence problem. The following subsections describe the theory behind each stage as it applies to the circuit implemented in this project.

1. Step-Down Transformer Stage

The 220V, 50Hz AC mains is fed into the primary winding of a center-tapped transformer rated 6V–0–6V (12V AC total) on the secondary side. Through electromagnetic induction, the mains voltage is stepped down to this safer AC level at the secondary winding, in proportion to the transformer's turns ratio ($N_p : N_s$). This isolation is critical for safety, since the secondary side is galvanically separated from the high-voltage mains, and it also provides a manageable voltage level for the downstream semiconductor devices, which are typically only rated to operate at a few tens of volts at most. The center tap additionally provides the reference (ground) point that the two-diode full-wave rectifier needs in order to operate.

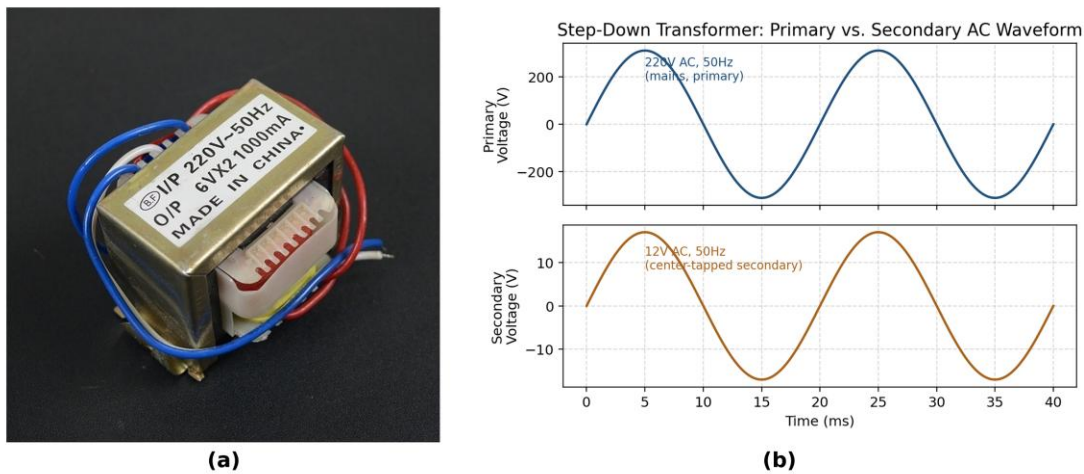


Fig. 1: (a) Photograph of the step-down transformer used in this project; (b) primary (220V) and secondary (12V, center-tapped) AC waveforms of the transformer stage

2. Full-Wave Rectification Stage

A two-diode full-wave rectifier (D1, D2) is connected across the center-tapped secondary winding, drawn from the five general-purpose diodes procured for the project (the remaining diodes were kept as tested spares). During the positive half-cycle, D1 is forward-biased and conducts current to the load; during the negative half-cycle, D2 becomes forward-biased instead. This ensures that current flows through the load in the same direction during both half-cycles, effectively doubling the ripple frequency to 100Hz and making it much easier to filter than a half-wave rectifier, whose output falls to zero for an entire half-cycle.

Two practical figures of merit are normally used to characterize a rectifier stage. The rectification efficiency, $\eta = P_{dc} / P_{ac}$, expresses what fraction of the AC input power is converted into usable DC output power; a full-wave rectifier has a theoretical maximum efficiency of about 81.2%, roughly double that of a half-wave rectifier (40.6%). The other important figure is the peak inverse

voltage (PIV) that each diode must withstand while reverse-biased for the two-diode, center-tapped topology used here, each diode must be rated for at least twice the peak secondary voltage of one half-winding, since the full secondary voltage appears across the non-conducting diode during the opposite half-cycle.

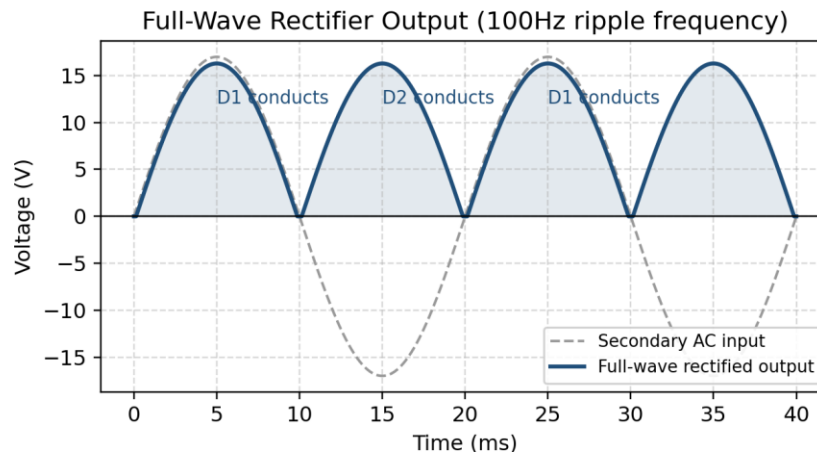


Fig. 2: Secondary AC input and the resulting full-wave rectified (pulsating DC) output, showing D1/D2 conduction intervals

3. Filter (Ripple Reduction) Stage

The output from the rectifier is a pulsating DC waveform — unidirectional, but far from constant. To smooth this out, a 1000 μF , 25–50V electrolytic capacitor (C1) is placed in parallel with the load, and it works in two simple steps that repeat every cycle:

- Charging: whenever the rectifier's output rises above the capacitor's own voltage, the conducting diode allows current through, and the capacitor quickly charges up to close to the peak voltage of the rectified wave.
- Discharging: once the rectified wave starts falling below the capacitor's voltage, both diodes turn off, and the capacitor slowly discharges through the load resistor instead, holding the output up rather than letting it fall all the way to zero.

This repeated charge–discharge action is what turns the sharp pulsating peaks from the rectifier into a much smoother, more constant DC level, as shown in Fig. 3. A larger capacitor discharges more slowly for the same load current, so it holds the voltage up longer between peaks and gives a smoother output — at the cost of a larger physical size and a higher inrush current at power-up. The 1000 μF value used in this project represents a practical compromise for the modest load currents (a few tens of milliamps) drawn in this circuit.

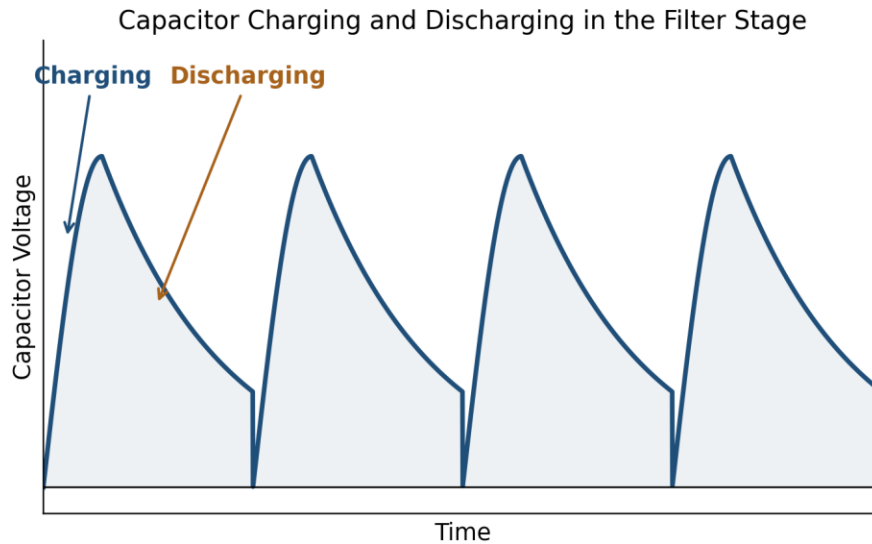


Fig. 3: Repeated charging and discharging of the filter capacitor C1, which smooths the rectifier's pulsating output

4. Voltage Regulation Stage (Stabilization)

Even with filtering, the DC voltage will still fluctuate if the load current changes or if the AC input varies, because the filter capacitor's ripple and average voltage both depend on the load. To combat this, a Zener diode rated $V_z = 6.2\text{V}$ is placed in parallel with the output, preceded by a 200Ω , 1W series resistor (R_1) that limits the current drawn from the filtered supply and drops the excess voltage. When reverse-biased beyond its Zener (breakdown) voltage, the diode operates in a region where a very large change in current produces only a very small change in voltage that is, it presents a low dynamic resistance (r_z) and it therefore maintains a nearly constant voltage across its terminals by dynamically adjusting the current passing through it. In effect, any current not drawn by the load is simply shunted through the Zener diode to ground, and vice versa, so that the sum of the load current and the Zener current supplied through R_1 remains essentially constant.

A $10\text{k}\Omega$ resistor provides a fixed bias/bleeder path across the output, ensuring the Zener always carries at least a small minimum current even under light-load conditions, and the variable load R_L is connected in parallel with it to study load regulation. A well-designed Zener regulator additionally depends on the series resistor R_1 being small enough that the Zener current never falls to zero (which would collapse the regulation) across the whole expected load range, yet large enough that the diode's maximum power rating is never exceeded when the load is removed and the full bias current flows through it alone.

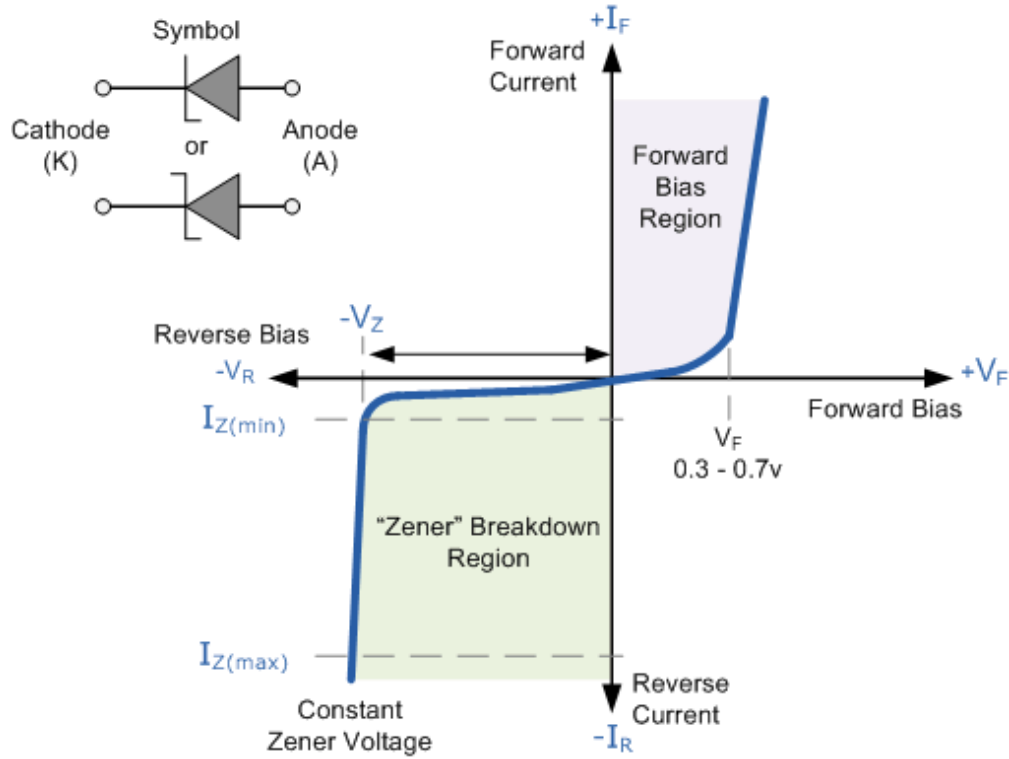


Fig. 4: Zener diode I-V characteristic

5. Design Calculations

A short hand calculation illustrates how R_1 was sized. Taking the filtered, unregulated DC input to the regulator as approximately $V_{in} \approx 9V$ (the peak of the rectified 6V-per-half secondary, less the diode forward drops and the capacitor's ripple sag), and requiring the Zener to carry at least a minimum current of $I_{Z(min)} \approx 5mA$ even at the heaviest tested load current of $I_{L(max)} \approx 12.17mA$, the series resistor is estimated from:

$$R_1 = \frac{V_{in} - V_Z}{I_{Z(min)} + I_{L(max)}} = \frac{9 - 6.2}{0.005 + 0.01216} \approx 163\Omega$$

The next standard, more conservative resistor value of 200Ω was selected, which keeps the Zener comfortably biased across the whole load range while adding a small safety margin. At no load, essentially the full bias current flows through the Zener diode alone; taking

$$I_{Z(max)} \approx \frac{V_{in} - V_Z}{R_1} \approx \frac{9 - 6.2}{200} \approx 14mA$$

the worst-case Zener power dissipation is approximately

$$P_Z = V_Z \times I_{Z(max)} \approx 6.2V \times 14mA \approx 87mW$$

well within the safe rating of a standard small-signal Zener diode. Similarly, the worst-case power dissipated in R_1 is

$$P_{(R1)} = I^2 R \approx (0.0172A)^2 \times 200\Omega \approx 59mW$$

comfortably inside the 1W rating chosen for R_1 and leaving a wide thermal safety margin one of the reasons a 1W resistor, rather than the more common $\frac{1}{4}W$ type, was specified for this position.

Therefore, the complete regulated power supply can be summarized as the following signal chain:

AC Mains (220V) → Transformer (12V AC, center-tapped) → Full-Wave Rectifier → Filter Capacitor (1000 μ F) → Series Resistor (R_1) → Zener Diode (6.2V) → Stable DC Output

Circuit Diagram

The complete schematic of the implemented power supply transformer, full-wave rectifier, filter capacitor, series resistor, Zener regulator, bias resistor and variable load is shown in Fig. 5. Every node in the physical breadboard layout corresponds directly to a labelled node in this diagram, which made it straightforward to probe intermediate test points (e.g., immediately after the rectifier, and again after the filter capacitor) with the multimeter and oscilloscope during construction and debugging.

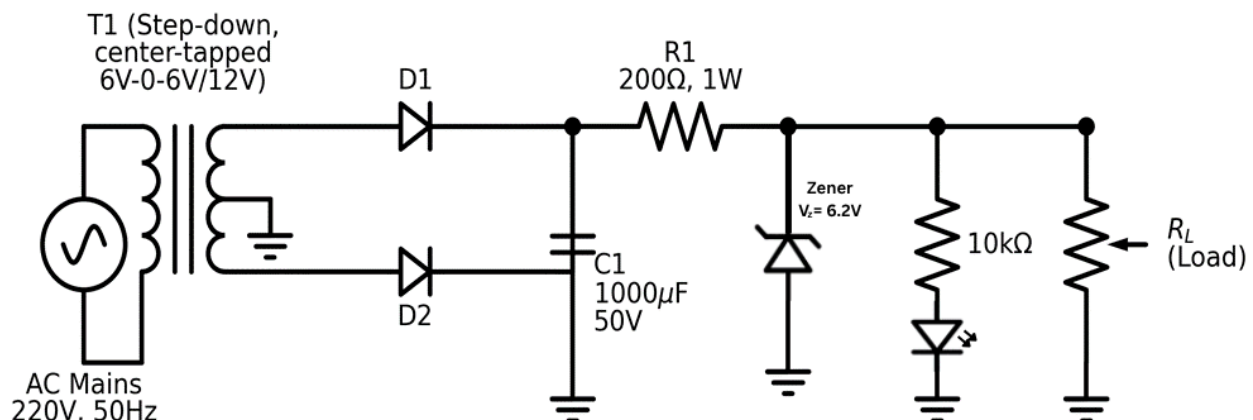


Fig. 5: Complete circuit diagram of the transformer–rectifier–filter–Zener regulated power supply

Components, Circuit Symbols and Functions

Table 1 lists every component used in the circuit together with its standard schematic symbol, the specific rating used in this project, and its function within the power supply.

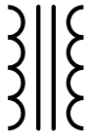
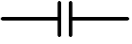
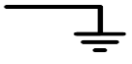
Component	Symbol	Rating Used	Function
Step-down Transformer		220V AC to 6V–12V AC, center-tapped (T1)	Steps down and electrically isolates the 220V mains to a safe secondary AC voltage.
Rectifier Diode		5 pcs (2 used as D1, D2; 3 kept as spares)	Converts AC into pulsating unidirectional DC during the full-wave rectification stage.
Electrolytic Capacitor		1000 μ F, 25V–50V (C1)	Filters the pulsating DC, storing charge to smooth the waveform and reduce ripple.
Zener Diode		$V_Z = 6.2V$	Provides a stable reference/regulated output voltage when reverse-biased in breakdown.
Fixed Resistor		200 Ω 1W (R1, series); 10k Ω (bias); 1k Ω , 2.2k Ω , 4.7k Ω , 10k Ω 6 pcs each	Limits Zener current, sets the output bias/bleeder path, and current-limits each LED branch.
Light Emitting Diode (LED)		6 pcs	Visually indicates a steady, flicker-free regulated output on each load branch.
Variable Load (RL)		Combinations of the 1k Ω –10k Ω resistors, 500 Ω – ∞	Emulates a variable load to study the load-regulation behaviour of the regulator.
Ground		Common reference (0V)	Common return path/reference node for the whole circuit.

Table 1: Components, symbols and functions used in the power supply

A few further remarks on the components are worth noting. The transformer is the only component that touches the mains side of the circuit directly, so its insulation and core rating were checked

carefully before use, and all mains-side wiring was fully taped and isolated. The five rectifier diodes procured for the project were all of the same general-purpose type; using only two of them in the rectifier bridge, with the rest set aside as tested spares, guarded against the possibility of a damaged or mismatched diode causing an asymmetric and therefore rippling rectified waveform. The 1000 μF filter capacitor is polarized, so its orientation was double-checked against the marked stripe before power-up, since reverse-biasing an electrolytic capacitor can cause it to fail catastrophically. Finally, each of the six LEDs was paired with its own resistor from the 1k Ω –10k Ω sets rather than being driven directly from the rail, both to limit its forward current to a safe value and to allow each branch to be tested independently.

Experimental Method

The project was implemented and tested step by step, following the sequence below, so that each stage of the circuit could be verified independently before the next stage was connected.

1. Component Collection

We gathered all the necessary components: a step-down center-tapped transformer (220V to 6V–12V AC), five rectifier diodes, a 1000 μF /25–50V electrolytic capacitor, a 6.2V Zener diode, resistors (200 Ω 1W, 10k Ω , and six each of 1k Ω , 2.2k Ω , 4.7k Ω and 10k Ω), six LEDs, a breadboard, and connecting wires. Each component was visually inspected and, where practical, checked with a multimeter (diode-test mode for the diodes and LEDs, resistance mode for the resistors) before being placed on the breadboard, to rule out any dead-on-arrival parts before assembly began.

2. Circuit Assembly Rectifier Stage

First, we connected an AC cable to the transformer, ensuring there were no loose connections. We also made sure never to touch the 220V AC supply by taping the exposed wires completely. The center-tapped secondary output of the transformer was connected to two of the five diodes (D1, D2) to form the full-wave rectifier. The anode sides were connected to the transformer secondary, and the rectifier output was checked with a multimeter and oscilloscope to confirm pulsating DC at roughly twice the mains frequency.

3. Filter Stage Connection

A 1000 μF electrolytic capacitor was connected across the output of the full-wave rectifier, observing the correct polarity. This reduced the ripple and produced a smoother DC waveform. The improvement was verified by measuring the DC voltage, and observing the waveform on the

oscilloscope, both with and without the capacitor connected, which made the ripple-smoothing effect of the capacitor directly visible.

4. Regulator Stage Connection

The filtered DC output was fed through the 200Ω , 1W series resistor (R_1) to the input of the 6.2V Zener diode. The regulator's output was measured and confirmed to be a stable 6.2V DC, with a $10k\Omega$ resistor providing a fixed bias/bleeder path across the output so that the Zener remained in regulation even before any external load was attached.

5. Load Testing

Six LEDs, each with its own current-limiting resistor (drawn from the $1k\Omega$, $2.2k\Omega$, $4.7k\Omega$ and $10k\Omega$ sets), were connected to the regulated output; all six glowed steadily, showing a stable voltage with no visible flicker even as neighbouring branches were connected and disconnected. The load (R_L) was then swept through twelve steps, from open-circuit down to the heaviest load available from the component set, using only the four stock resistor values on hand the single values ($10k\Omega$, $4.7k\Omega$, $2.2k\Omega$, $1k\Omega$) directly, and every intermediate resistance built by combining two of them in series or parallel down to two $1k\Omega$ resistors in parallel for the heaviest 500Ω load. An ammeter was connected in series with the load to measure the load current (I_L), and the output voltage (V_o) was recorded with a digital multimeter for each step.

6. Observation and Recording

The output voltage before and after filtering was compared, and the ripple waveform was sketched from the oscilloscope trace. Voltage regulation under the twelve different load conditions was recorded in a data table. Performance was then analyzed by comparing the theoretical expectations developed in the Theory section against the experimental results, and any discrepancies were noted for discussion.

7. Safety Precautions

Proper insulation was maintained while working with the 220V AC mains, and the circuit was only powered on after every mains-side connection had been double-checked. All components were tested with a multimeter before use, and the circuit was assembled and re-checked with the power off before each new stage was energized. Components were also handled carefully to prevent overheating, particularly the R_1 series resistor and the Zener diode, which dissipate the most excess power, and the breadboard area around them was kept clear to allow air circulation.

Equipment and Components Needed

- Transformer (220V to 6V–12V AC, center-tapped): steps down the mains AC supply to a safer low voltage.
- Rectifier Diodes (5 pcs): used to rectify AC into pulsating DC (two used in the full-wave pair).
- Capacitor (1000 μ F, 25V–50V Electrolytic): used as a filter to smoothen the rectified output.
- Zener Diode ($V_z = 6.2\text{V}$): provides a regulated 6.2V DC constant output.
- Resistor (200 Ω , 1W): series current-limiting resistor ahead of the Zener diode.
- Resistor (10k Ω): fixed bias/bleeder resistor across the regulated output.
- Resistors (1k Ω , 2.2k Ω , 4.7k Ω , 10k Ω 6 pcs each): LED current limiting and variable-load (R_L) test resistors.
- LEDs (6 pcs): to test the output visually.
- Breadboard and Wires: for circuit assembly.
- Plug / AC cable: for the mains power connection.
- Digital multimeter and oscilloscope: for voltage, current and waveform measurement.

Data Table

With the Zener regulator set at $V_z = 6.2\text{V}$, the load resistance was stepped from open-circuit (no load) down to 500 Ω , and the load current and output voltage were recorded at each step.

Obs. No.	Load Resistance (R_L in k Ω)	Load Current (I_L in mA)	Output Voltage (V_o in V)
1	∞ (No Load)	0.000	6.128
2	10	0.613	6.126
3	9.4	0.652	6.126
4	6.9	0.888	6.125
6	5.7	1.074	6.124
7	4.7	1.303	6.123
8	4.4	1.392	6.123
9	3.2	1.914	6.121
10	2.2	2.781	6.118
11	1	6.106	6.106
12	0.5	12.167	6.084

Table 2: Load regulation data for the 6.2V Zener-regulated power supply, sorted by load resistance

Graph

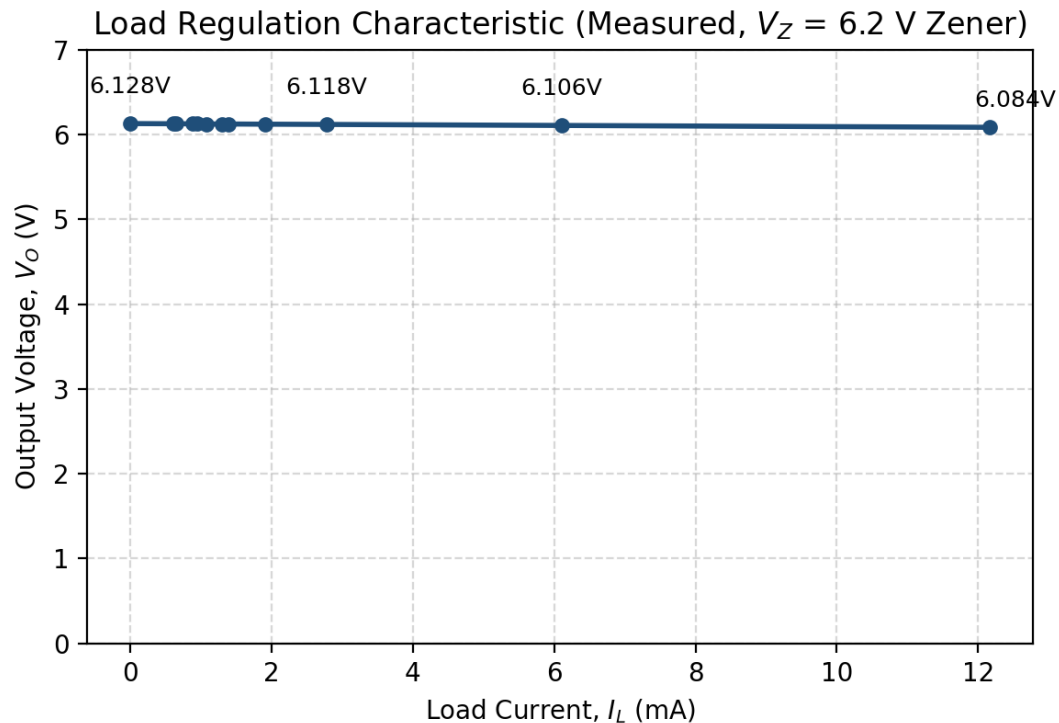


Fig. 6: Load regulation characteristic (V_o vs. I_L) of the 6.2V regulated power supply

From the no-load and full-load readings, the percentage load regulation of the circuit is calculated as:

$$\% \text{ Load Regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100 = \frac{6.128 - 6.084}{6.084} \times 100 \approx 0.723\%$$

A load regulation figure of well under 1% indicates that the Zener regulator holds the output within a very tight band across the full tested load-current range (0 to over 12mA), consistent with the very low dynamic resistance expected of a Zener diode operated comfortably above its knee current. This is noticeably tighter than the ripple seen between individual readings, which is dominated by the ± 0.01 – 0.02 V resolution of the multimeter rather than by any real change in the circuit's behavior.

Results

The full-wave rectifier successfully converted AC to DC, but the output contained ripples clearly visible on the oscilloscope. With the 1000 μF capacitor filter, the ripple voltage was significantly reduced, and the waveform flattened into a near-horizontal line with only a small residual sawtooth ripple riding on top of the average DC level. The Zener-regulated output gave a near-constant 6.1V-class DC level across the tested load range, drooping only from 6.128V at no load to 6.084V at 12.167mA a load regulation of about 0.72%. All six LEDs glowed uniformly with no visible flicker across the tested load range, confirming that the regulator could comfortably supply the combined LED current alongside the variable test load. The overall efficiency of the circuit was satisfactory for small-scale, low-current applications, though as expected for a linear regulator a portion of the input power was necessarily dissipated as heat in R_1 and the Zener diode rather than delivered to the load.

Discussion

The project clearly illustrates the critical role of each stage in a regulated power supply. The center-tapped transformer safely stepped down the 220V AC mains voltage to a lower, manageable AC voltage suitable for electronic circuits, highlighting the importance of isolation and voltage-level compatibility. The two-diode full-wave rectifier efficiently converted AC to pulsating DC, utilizing both half-cycles and demonstrating superior performance both in ripple frequency and in rectification efficiency compared to a half-wave rectifier. The 1000 μF capacitor filter significantly smoothed the pulsating DC, reducing voltage ripple and ensuring that sensitive components downstream could operate reliably.

Observations during testing aligned closely with theoretical predictions, validating the principles of rectification, filtering and Zener regulation covered in the Theory section. However, practical limitations were also evident: the forward voltage drop across the diodes reduced the peak rectified voltage below the ideal value, and the power dissipated in the 200 Ω series resistor and the Zener diode itself slightly reduced the overall efficiency of the circuit. The output voltage also drooped gradually as the load current increased toward 12.17mA, rather than remaining perfectly flat, because the Zener diode's small but non-zero dynamic resistance causes its terminal voltage to fall slightly as more current is diverted away from it and into the load.

Several practical sources of error were also identified. Tolerances in the resistor values (typically $\pm 5\%$) and in the Zener voltage itself (commonly $\pm 5\%$ around the nominal 6.2V rating) mean that the measured values will not exactly match a hand calculation performed with nominal component

values which is consistent with the measured no-load output settling near 6.13V rather than exactly 6.20V. Contact resistance at the breadboard connections, and the internal resistance of the ammeter used to measure I_L , both introduce small additional voltage drops that were not accounted for in the idealized theory. The raw readings also scattered by ± 0.01 – 0.02 V from one observation to the next with no consistent trend, which is best explained by the least-count resolution of the digital multimeter rather than any real change in the circuit; fitting a straight line to V_o against I_L , as shown in the Data Table section, separates this measurement noise from the genuine, physically expected droop caused by the Zener diode's small dynamic resistance. Mains-voltage fluctuations during testing could likewise have caused small, otherwise unexplained shifts in the readings between observations.

These real-world effects emphasized the importance of considering component characteristics, tolerances, and thermal management in power-supply design, rather than relying solely on idealized textbook equations. For higher-current applications, a heat sink on the series resistor, a higher-power Zener diode, or a switching regulator in place of the simple linear stage could all enhance performance, highlighting the trade-offs between simplicity, cost, reliability, and efficiency that a designer must weigh for any given application.

Applications of Regulated DC Power Supplies

Linear, Zener-regulated power supplies of the type built in this project are widely used wherever a small, steady DC voltage is needed and the load current is modest for example, to bias sensor circuits, to power small microcontroller boards and logic-level ICs, to drive indicator LEDs and low-power relays, and as a simple benchtop reference-voltage source in electronics laboratories. Because the design is inexpensive, easy to build, and produces very low output noise compared to a switching regulator, it remains a common building block in low-power analog and mixed-signal circuits, even though switching regulators are generally preferred for higher-current or battery-powered applications where efficiency is critical.

Advantages and Limitations of the Circuit

The principal advantages of the design implemented here are its simplicity, its low component count, its low output ripple and noise, and its very fast transient response since the Zener diode reacts to load changes essentially instantaneously, without the switching delay and output ripple associated with switch-mode regulators. It is also inexpensive and straightforward to troubleshoot with basic laboratory instruments, which makes it well suited to an introductory circuits course.

Its main limitations are its relatively low efficiency at light load since the bias current through R1 and the Zener diode is wasted whenever the load draws less than the full bias current and its limited output-current capability, since increasing the load current beyond the design range would either starve the Zener diode of bias current (collapsing the regulation) or force the use of a bulkier, higher-power series resistor and Zener diode. The design is also sensitive to the input voltage: if the transformer's secondary voltage sags under heavy mains loading, the regulator may lose regulation entirely once V_{in} falls close to V_z . For applications needing higher current, better efficiency, or a well-defined variable output voltage, a three-terminal linear IC regulator (such as the 7805/7806 family) or a switching regulator would typically be preferred over the discrete Zener-diode approach used in this project.

Conclusion

The regulated power supply was successfully designed, implemented, and tested, achieving all of the project's stated objectives. Starting from a 220V AC mains connection, the circuit progressively stepped the voltage down through a center-tapped transformer, rectified it with a two-diode full-wave rectifier, smoothed it with a 1000 μF filter capacitor, and finally stabilized it with a nominal 6.2V Zener regulator biased through a carefully sized 200 Ω series resistor. The system converted high-voltage AC mains into a stable, approximately 6.1V DC output, and performance remained reliable with a measured load regulation of only about 0.72% across the entire tested range of load conditions, from no load up to a load current of 12.167mA.

This project offered substantial practical insight into essential electronic components, including transformers, rectifier diodes, filter capacitors, Zener regulators, resistors and LEDs, and reinforced the theory covered in lectures with hands-on measurement and analysis. Building the circuit stage by stage, and verifying each stage independently with a multimeter and oscilloscope before moving on to the next, proved to be an effective and disciplined approach to a multi-stage analog design, and it made isolating the small discrepancies between the theoretical and measured results such as diode forward-voltage drops, resistor tolerances, and the Zener diode's non-zero dynamic resistance far easier than debugging the completed circuit all at once.

The hands-on experience also allowed direct observation of real-world engineering challenges that are easy to overlook in a purely theoretical treatment, such as voltage drops across seemingly ideal components, residual ripple that never quite reaches zero, and heat dissipation in the regulation stage, all of which emphasize the need for careful component selection, appropriate power ratings, and thermal management in any practical power-supply design. Working through the accompanying hand calculations for R1 and then checking those calculations against the measured

Zener current and power dissipation also demonstrated how approximate, first-pass design equations are used in practice to select standard component values with an appropriate safety margin, rather than to predict an exact numerical outcome.

Looking forward, several extensions could meaningfully improve this design. Switching to a monolithic linear regulator IC or a switching regulator topology would improve efficiency and current-handling capability considerably; integrating a small digital voltmeter or LED bar-graph display would allow the output to be monitored in real time without external test equipment; and adding a second, independently regulated output rail would let the same transformer and rectifier stage serve dual-voltage applications, such as circuits requiring both 3.3V and 5V logic supplies. Overall, the project not only confirmed the theoretical knowledge covered in the CSE 251 coursework but also meaningfully enhanced practical skills in circuit assembly, measurement, troubleshooting, and design justification, providing a comprehensive, end-to-end understanding of regulated power-supply design.